

Instructions: Fill in the blanks.

1. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, $\frac{(n-1)S^2}{\sigma^2} \sim$ _____.

Solution: χ_{n-1}^2

2. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, $\bar{Y} \sim$ _____.

Solution: $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

3. (2 pts) Let $W_1 \sim \chi_{\nu_1}^2$ and $W_2 \sim \chi_{\nu_2}^2$ be independent RVs. Then,

$$F = \frac{W_1/\nu_1}{W_2/\nu_2} \sim F_{\nu_1, \nu_2}.$$

Solution: $\frac{W_1/\nu_1}{W_2/\nu_2}$

4. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, $\bar{Y}$ and $S^2 =$ _____ are
_____.

Solution: $\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$; independent

5. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z_i = \frac{Y_i - \mu}{\sigma}$. Then, $\sum_{i=1}^n Z_i^2 \sim$ _____.

Solution: χ_n^2

6. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \frac{\bar{Y} - \mu}{S/\sqrt{n}}$. Then, $T \sim$ _____.

Solution: t_{n-1}

7. (2 pts) Suppose $Z \sim N(0, 1)$ and $W \sim \chi_\nu^2$. Then, if Z and W are independent,

$T =$ _____ $\sim t_\nu$.

Solution: $\frac{Z}{\sqrt{W/\nu}}$

8. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z_i = \frac{Y_i - \mu}{\sigma}$. Then, $Z_i^2 \sim$ _____.

Solution: χ_1^2

9. (2 pts) Suppose $Y \sim \text{Gamma}(\alpha = \frac{n}{2}, \beta)$, where n is a positive integer.

Let $W =$ _____. Then, $W \sim \chi_n^2$.

Solution: $\frac{2Y}{\beta}$

10. (3 pts) **Central Limit Theorem:** Let $Y_1, Y_2, \dots, Y_n$ be i.i.d. RVs with $E(Y_i) = \mu$ and

$V(Y_i) = \sigma^2 < \infty$. Then, as _____,

_____ converges in distribution to _____.

Solution: $n \rightarrow \infty$; $\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$; $N(0, 1)$

11. (2 pts) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$. Then, $Z \sim$ _____.

Solution: $N(0, 1)$

12. (2 pts) Suppose $Y_1, \dots, Y_n \stackrel{indep.}{\sim} \text{Gamma}(\alpha_i, \beta)$. Let $U = \sum_{i=1}^n Y_i$. Then,

$U \sim$ _____.

Solution: $\text{Gamma}\left(\sum_{i=1}^n \alpha_i, \beta\right)$